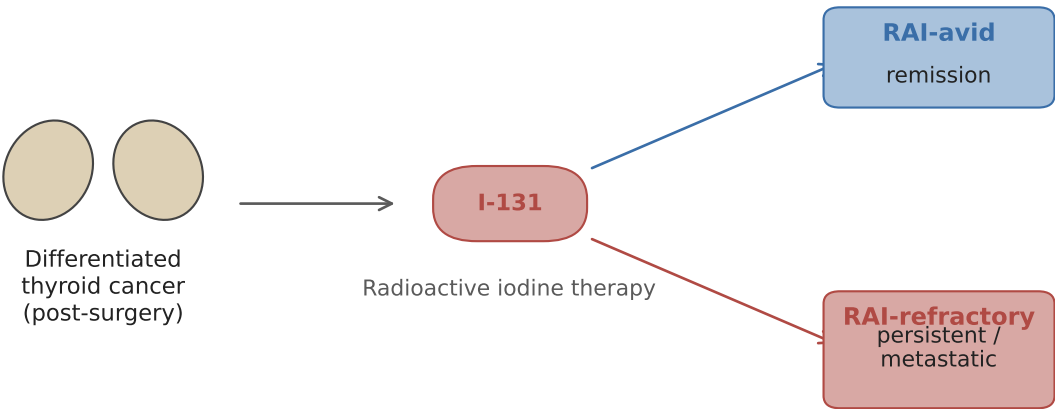


A multi-gate model of radioiodine failure in thyroid cancer

a

Clinical unmet need

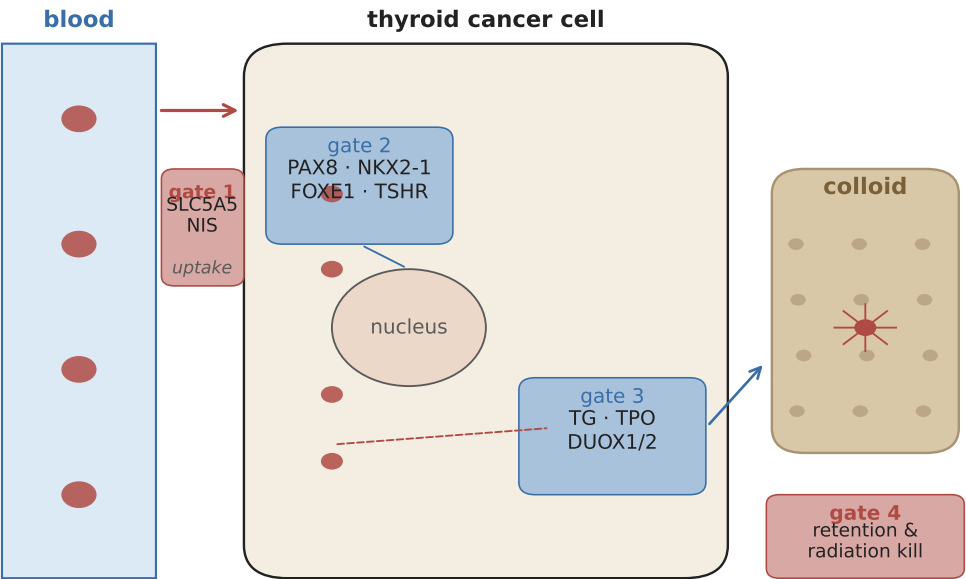


10-yr survival drops > 90% (RAI-responsive) → ~10–14% (distant RAI-R-DTC)

↓ a parsimonious molecular readout to triage RAI candidates

b

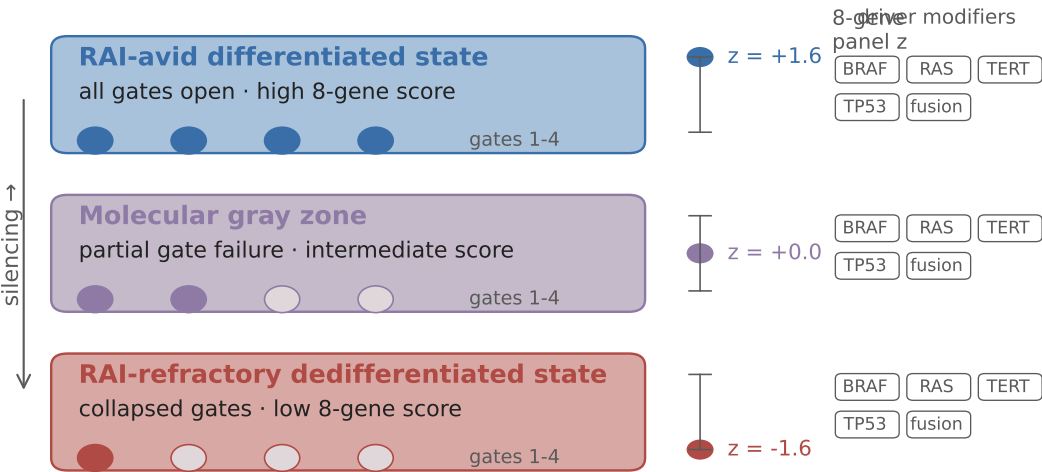
Multi-gate biology of RAI therapy



NIS alone is insufficient — coordinated thyroid differentiation across all four gates is required.

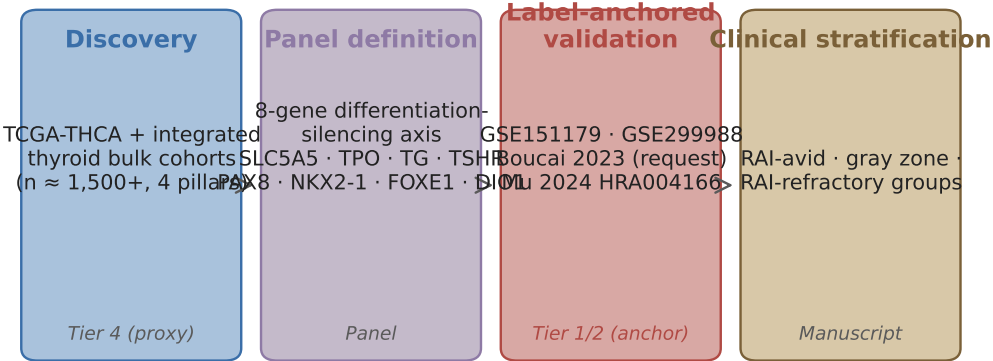
c

Three molecular states along the differentiation-silencing axis



d

Evidence ladder



Tier 4 proxy = discovery context (no direct RAI labels in TCGA).
Tier 1/2 anchor = RECIST or RAI-avidity labels in independent cohorts.